u_media

media_engine

busy

clk

cs_n
dim_half

f_en

frame_tick has_snd

j_en miso mosi

pic_ok

play

ready

sclk
see

spr_ok
spr_on

stop

atk_first[4:0]
f_addr[11:0]

from_rec[15:0]
item[4:0]
j_addr[14:0]

lose_first[4:0]

lose_n[4:0] mode[1:0] mus_first[4:0] mus_n[4:0] pcm[15:0]
rec_pos[15:0] rgb[11:0]

spr_rgb[11:0] win_first[4:0] win_n[4:0]

x[9:0] y[9:0]

zone[1:0]

u_spi

spi_flash

sclk

btn_hard

btn_left btn_right btn_rot btn_soft

buzzer

c_left_i RTL_MUX

I0 S=1'b1

I1 S=default O S

c_right_i RTL_MUX

I0 S=1'b1

I1 S=default O S

chaos_en_i RTL_AND

I0

I1 O

clk

dig[3:0]

f_en_i RTL_AND

I0

I1 O

flash_hold_n

flash_cs_n

flash_miso

flash_wp_n

flash_mosi

g_hard_i RTL_AND

I0

I1 O

g_left_i RTL_AND

I0

I1 O

g_left0_i RTL_INV

I0 O

g_right_i RTL_AND

I0

I1 O

g_rot_i RTL_AND

I0

I1 O

g_soft_i RTL_AND

I0

I1 O

g_startup.u_startup

STARTUPE2

CFGCLK

n/c CFGMCLK
n/c EOS
n/c PREQ

n/c

CLK GSR GTS

KEYCLEARB

PACK USRCCLKO

USRCCLKTS USRDONEO
USRDONETS

garb_in_i RTL_OR

I0

I1 O

garb_in_n_i RTL_MUX S

I0[4:0] S=1'b1

I1[4:0] V=B"00011", S=default O[4:0]

g0_i RTL_MUX S

I0[6:0] S=1'b1

I1[6:0] S=default O[6:0]

g1_i RTL_MUX S

I0[6:0] S=1'b1

I1[6:0] S=default O[6:0]

g2_i RTL_MUX S

I0[6:0] V=B"1000000", S=1'b1

I1[6:0] S=default O[6:0]

g3_i RTL_MUX S

I0[6:0] V=B"1000000", S=1'b1

I1[6:0] S=default O[6:0]

g31_i RTL_AND

I0

I1 O

hex7_i RTL_ROM

A[3:0] O[6:0]

hex7_i__0 RTL_ROM

A[3:0] O[6:0] hex7_i__1 RTL_ROM

A[3:0] O[6:0]

hex7_i__2 RTL_ROM

A[3:0] O[6:0] hex7_i__3 RTL_ROM
A[3:0] O[6:0] hex7_i__4 RTL_ROM

A[3:0] O[6:0]

jev_i RTL_AND

I0

I1 O

jev0_i RTL_AND

I0

I1 O

link_rx

link_tx

n_i /RTL_DIV

I0[4:0]

I1[3:0] V=B"1010" O[1:0]

n_i__0 %
RTL_MOD

I0[4:0]

I1[3:0] V=B"1010" O[3:0]

rx_line_i RTL_MUX

I0 S=1'b1

I1 S=default O S

seg_n[7:0]

show_i RTL_AND

I0

I1 O

sw_chaos

sw_music

sw_pause

sw_sfx

sw_task[7:0] sw_usb

touch_i RTL_OR

I0

I1 O
touch0_i RTL_OR

I0

I1 O

touch1_i RTL_OR

I0

I1 O

touch2_i RTL_OR

I0

I1 O

u_core

tetris_core

act_hard

act_left act_right act_rot act_soft

can_start clear_evt

clk
frame_tick
fx_freeze
fx_next_bad

fx_next_i fx_sink
fx_speed

game_over

garb_in

garb_out

lost_out opp_lost pause

playing

start_in

start_out

versus

clear_n[2:0] colf[0][29:0] colf[1][29:0] colf[2][29:0] colf[3][29:0] colf[4][29:0] colf[5][29:0] colf[6][29:0] colf[7][29:0] colf[8][29:0] colf[9][29:0] colf[10][29:0] colf[11][29:0] colf[12][29:0] colf[13][29:0] colf[14][29:0] colf[15][29:0] colf[16][29:0] colf[17][29:0] colf[18][29:0] colf[19][29:0] cur_color[2:0] cur_shape[15:0] cur_x[4:0] cur_y[5:0]

garb_in_n[4:0]

garb_out_n[4:0] ghost_y[5:0] height[4:0]

level[3:0]

lines_bcd[15:0] next_color[2:0] next_shape[15:0] occ[0][9:0] occ[1][9:0] occ[2][9:0] occ[3][9:0] occ[4][9:0] occ[5][9:0] occ[6][9:0] occ[7][9:0] occ[8][9:0] occ[9][9:0] occ[10][9:0] occ[11][9:0] occ[12][9:0] occ[13][9:0] occ[14][9:0] occ[15][9:0] occ[16][9:0] occ[17][9:0] occ[18][9:0] occ[19][9:0] pending[4:0]

result[2:0]

u_crender

chaos_render

banner_on
bounce

clk enable

face_en

frame_tick
locked

ov_on

t_active

banner_id[4:0]

face_addr[11:0]

n_bounce[3:0]

ov_rgb[11:0]

t_count[3:0]
t_meter[7:0]
t_offsecs[2:0]
t_progress[7:0]
t_secs[4:0]
t_sel[2:0]
t_sw[7:0]
t_target[7:0]
t_task[2:0]
t_zone_hi[7:0]

t_zone_lo[7:0] x[9:0] y[9:0]

u_fx

chaos_fx

banner_on

blind bonus bounce

clk enable

flip

frame_tick

freeze

garbage
got_attack hit_evt

locked media_hit mirror next_bad

next_i

paused
sent_attack

sink

speed

task_fail task_ok

banner_id[4:0]

bonus_id[1:0]

got_id[4:0] media_id[1:0] n_bounce[3:0] sent_id[3:0]

u_input

tetris_input

act_hard

act_left act_right act_rot act_soft

clk
frame_tick

btn[4:0]

u_link

versus_link

clk
frame_tick

got_attack got_garbage
got_lost got_start

link_ok

rx

send_attack
send_garbage

send_lost send_start

tx

attack_id[4:0]

garbage_n[4:0]

got_attack_id[4:0]

got_n[4:0] my_height[4:0]

opp_height[4:0]

u_mctrl

media_ctrl

busy

clk has_snd

jevil media_hit play

playing
ready

show

stop won

atk_first[4:0]

from_rec[15:0]
item[4:0]

lose_first[4:0]

lose_n[4:0] media_id[1:0] mus_first[4:0] mus_n[4:0]

rec_pos[15:0] result[2:0] win_first[4:0] win_n[4:0] vga_r[3:0] vga_vs

sclk_r_reg

RTL_REG C

CED Q

u_mix

media_mix

clk dim_half
frame_tick

g_hs g_vs

j_en jevil keep m_see
s_on

see_bg
show

vga_hs

vga_vs

won

g_b[3:0] g_g[3:0] g_r[3:0]

j_addr[14:0] m_rgb[11:0] m_zone[1:0] mode[1:0] s_rgb[11:0]

vga_b[3:0] vga_g[3:0] vga_r[3:0]

x[9:0] y[9:0]

u_render

tetris_render

can_start clk

fx_blind
fx_flip

game_over hsync keep

link_ok

ov_on

paused

playing

see_bg

vga_hs

vga_vs

video_on
vsync

colf[0][29:0]

colf[10][29:0] colf[11][29:0] colf[12][29:0] colf[13][29:0] colf[14][29:0] colf[15][29:0] colf[16][29:0] colf[17][29:0] colf[18][29:0] colf[19][29:0]

colf[1][29:0] colf[2][29:0] colf[3][29:0] colf[4][29:0] colf[5][29:0] colf[6][29:0] colf[7][29:0] colf[8][29:0] colf[9][29:0]

cur_color[2:0] cur_shape[15:0] cur_x[4:0] cur_y[5:0]
ghost_y[5:0]

level[3:0]

lines_bcd[15:0] next_color[2:0] next_shape[15:0] occ[0][9:0]

occ[10][9:0] occ[11][9:0] occ[12][9:0] occ[13][9:0] occ[14][9:0] occ[15][9:0] occ[16][9:0] occ[17][9:0] occ[18][9:0] occ[19][9:0]

occ[1][9:0] occ[2][9:0] occ[3][9:0] occ[4][9:0] occ[5][9:0] occ[6][9:0] occ[7][9:0] occ[8][9:0] occ[9][9:0]

opp_height[4:0] ov_rgb[11:0] pending[4:0]

result[2:0]

vga_b[3:0] vga_g[3:0] vga_r[3:0]

x[9:0] y[9:0]

u_seg

seg7_scan

clk d0[6:0] d1[6:0] d2[6:0] d3[6:0]

dig[3:0] dp[3:0] seg_n[7:0]

u_sound

tetris_sound

audio
buzz

clear_evt clk

frame_tick garb_evt hit_evt music_on
playing

sent_evt sfx_on
task_fail_evt
task_ok_evt
task_start_evt

clear_n[2:0]

pcm[15:0]

result[2:0]

u_sw

switch_filter

clk

sw_in[12:0] sw_out[12:0]

u_sync

vga_sync

clk

frame_tick hsync
video_on
vsync

x[9:0] y[9:0]

u_task

chaos_task

active

bonus

clk enable

frame_tick hurry media_ok paused

send_attack

task_fail task_ok
task_start

touch

versus

attack_id[3:0] bonus_id[1:0] count[3:0] hands_secs[2:0] meter_pos[7:0] progress[7:0] secs[4:0] sel[2:0] sw[7:0]

target[7:0]
task[2:0]

zone_hi[7:0] zone_lo[7:0]

usb_rx

vga_b[3:0] vga_g[3:0] vga_hs

audio_l audio_r

usb_tx